

Appendix K: Vendors

Appendix K: Vendor and Supplier Directory

K.1 Linear Motion Components

K.1.1 Ball Screws and Linear Guides

Vendor	Products	Region	Website	Notes
THK	Linear guides, ball screws, actuators	Global	thk.com	Premium quality, OEM standard
HIWIN	Linear guides, ball screws, ballscrews	Global	hiwin.com	Good quality, competitive pricing
TBI Motion	Ball screws, linear guides	Asia/Global	tbimotion.com	Budget-friendly, decent quality
PMI (AMT)	Linear guides, ball screws	Taiwan/Global	pmi-amt.com	Mid-range quality and price
Schneeberger	Linear guides, ball screws (ground)	Europe/USA	schneeberger.com	Ultra-precision, high cost
SBC Linear	Budget linear guides, SBR rails	China/Global	sbclinear.com	Budget CNC, DIY-friendly

Recommendation: - **Precision CNC:** THK or HIWIN HG-series (HGH blocks, C5 ball screws) - **Budget CNC:** TBI Motion or PMI (C7 ball screws) - **DIY/Hobbyist:** SBC Linear supported rails, rolled ball screws

K.1.2 Rack and Pinion Systems

Vendor	Products	Region	Notes
Atlanta Drive Systems	Precision racks, helical racks	USA/Global	Wide module range (M1-M12)
Ozgear	Ground racks, pinions	Australia	High precision, CNC-ready
C&U	Standard spur racks	China/Global	Budget option, plasma tables

K.2 Motors and Drives

K.2.1 Stepper Motors and Drivers

Vendor	Products	Region	Website	Notes
Nema23.com	NEMA 17/23/34 steppers + drivers	USA	nema23.com	CNC-focused bundles, good support
StepperOnline (OOZOOO)	NEMA 11-42 steppers, closed-loop	Global	omc-stepperonline.com	Wide range, competitive pricing
Leadshine	Steppers, drivers (DM series)	China/Global	leadshine.com	Reliable drivers, industrial quality
Gecko Drive	G203V, G540 (4-axis controller)	USA	geckodrive.com	High-performance stepper drivers
ClearPath	Servo-steppers (hybrid closed-loop)	USA	teknico.com	No tuning required, step/dir input

Recommendation: - **Budget:** StepperOnline NEMA 23 + DM542 driver - **Mid-range:** Leadshine DM856 or Nema23.com bundles - **High-performance:** Gecko G203V or ClearPath SDSK

K.2.2 Servo Motors and Drives

Vendor	Products	Region	Notes
Yaskawa	Sigma-7 servo systems	Global	Industrial standard, EtherCAT
Mitsubishi	MR-J4 servo amplifiers	Global	High performance, expensive
Delta Electronics	ASDA-A2 servo drives	Asia/Global	Mid-range price, good quality
JMC	AC servo motors + iHSV drivers	China	Budget servo option for DIY
Teknic	ClearPath servos (step/dir interface)	USA	Easy setup, integrated driver

K.3 Spindles and VFDs

K.3.1 Spindle Motors

Vendor	Products	Power Range	Website	Notes
HQD (Hanqi)	Air-cooled, water-cooled spindles	0.8-7.5 kW	hqd-motor.com	Popular for DIY CNC, affordable
Jianken	ER11/16/20/25 spindles	1.5-5.5 kW	jkongmotor.com	Good quality, ISO20/30 tapers

Vendor	Products	Power Range	Website	Notes
ELTE	High-frequency spindles (24k RPM)	2.2-9 kW	eltemotor.com	Precision machining, engraving
Teknomotor	ISO/BT spindles, ATC-ready	3-20 kW	teknomotor.com	Industrial, tool changer compatible
Perske	Ultra-high-speed spindles	5-50 kW	perske.com	40,000+ RPM, very expensive

Recommendation: - **Router/Wood:** HQD 2.2 kW air-cooled (ER20 collet) - **Milling/Aluminum:** Jianken 3.0 kW water-cooled (ER25) - **High-speed:** ELTE 4.0 kW (24,000 RPM, ER20)

K.3.2 VFDs (Variable Frequency Drives)

Vendor	Models	Power Range	Notes
Huanyang	HY-series	0.75-7.5 kW	Budget VFD, RS485 Modbus, common in DIY
Teco	L510 series	0.75-15 kW	Reliable, industrial-grade
Delta Electronics	VFD-E series	0.4-3.7 kW	Compact, easy setup
Fuling	DZB200	1.5-5.5 kW	Mid-range price, good quality
ABB	ACS580	0.75-250 kW	Industrial, very reliable, expensive

K.4 Control Electronics

K.4.1 CNC Controllers

Controller	Interface	Axes	Software	Price	Best For
LinuxCNC	PC + Mesa FPGA card	Unlimited	Open-source	\$\$	Advanced users, custom machines
Mach3	PC + parallel port/USB	6	Windows (legacy)	\$	Hobbyist, older PCs
Mach4	PC + Ethernet/USB	6+	Windows (modern)	\$\$	Hobbyist/pro, modern hardware
UCCNC	PC + UC series controllers	6	Windows	\$\$	User-friendly, good support
Centroid Acorn	Standalone + PC	4-6	Embedded Linux	\$\$\$	Retrofit, industrial-grade
PathPilot	Dedicated controller	3-4	Embedded Linux	\$\$\$\$	Tormach machines, easy to use

Recommendation: - **Beginner:** Mach4 + ESS Ethernet controller - **Advanced:** LinuxCNC + Mesa 7i96 FPGA - **Plug-and-play:** Centroid Acorn or UCCNC

K.4.2 Breakout Boards (BOBs)

Vendor	Model	Axes	Inputs/Outputs	Interface	Notes
CNC4PC	C10, C11	4-5	10-20 I/O	Parallel port	Opto-isolated, charge pump Mach3/4 compatible, reliable LinuxCNC FPGA, advanced Industrial-quality, UL-listed UCCNC compatible, plug-and-play
Ethernet Smooth-Stepper	ESS	6	16 I/O	Ethernet	
Mesa Electronics	7i96	5	32 I/O	Ethernet	
PMDX	126, 134	4	8-16 I/O	Parallel/USB	
UC100	USB controller	6	16 I/O	USB	

K.5 Structural Materials

K.5.1 Aluminum Extrusion

Vendor	Products	Region	Website	Notes
80/20 Inc.	T-slot extrusion (10-series, 15-series)	USA	8020.net	Premium, USA-made, excellent selection
Misumi	HFS-series extrusion	Global	us.misumi-ec.com	Japanese quality, fast shipping
Faztek	T-slot framing systems	USA	faztek.com	Competitive pricing, large selection
Tnutz	Budget T-slot extrusion	USA	tnutz.com	Lower cost than 80/20, decent quality
Maytec	Aluminum profiles (40x80, 45x90)	Europe	maytec.com	European standard, metric sizing

Common Sizes: - **Light CNC:** 40x40mm, 40x80mm - **Medium CNC:** 60x60mm, 80x40mm - **Heavy CNC:** 80x80mm, 80x160mm

K.5.2 Steel and Plate

Vendor	Products	Region	Notes
Online Metals	Sheet, plate, bar stock	USA	onlinemetals.com
McMaster-Carr	All materials, fasteners, tools	USA	mcmaster.com
Speedy Metals	Precision ground plate, shafting	USA	speedymetals.com
Metal Supermarkets	Cut-to-size service	USA/Canada	metalsupermarkets.com

K.6 Plasma and Laser Components

K.6.1 Plasma Torches and Consumables

Vendor	Products	Notes
Hypertherm	Powermax series (45-125A)	Industry standard, expensive consumables
Everlast	PowerPro series (50-80A)	Budget-friendly, good for DIY tables
Razorweld	Cut series (45-60A)	Australian brand, reliable
Consumable parts:	Tips, electrodes, shields, nozzles	Use OEM or quality aftermarket (avoid cheap eBay)

K.6.2 Fiber Laser Sources and Optics

Vendor	Products	Power Range	Notes
Raycus	Fiber laser sources	20W-100kW	Chinese, budget-friendly, widely used
JPT	MOPA fiber lasers	20W-100W	Adjustable pulse, marking applications
IPG Photonics	Fiber lasers	100W-100kW	Premium, industrial-grade, very reliable
II-VI (Coherent)	CO ₂ lasers, cutting heads	40W-6kW	CO ₂ legacy, excellent optics

K.7 Waterjet Components

Vendor	Products	Notes
KMT Waterjet	Pumps, cutting heads (60k-90k PSI)	Industry leader, expensive
OMAX	Complete waterjet systems	USA-made, software integrated
Jet Edge	Ultra-high-pressure pumps (up to 94k PSI)	Custom systems, industrial

Vendor	Products	Notes
Waterjet Corporation	Abrasive delivery, nozzles, consumables	OEM and aftermarket parts

Abrasive Suppliers: - **Barton Garnet:** 80-mesh garnet (standard for waterjet) - **GMA Garnet:** Australian garnet, high quality

K.8 Electronics and Sensors

K.8.1 Limit Switches and Proximity Sensors

Vendor	Products	Notes
Omron	D2F micro switches, E2E proximity sensors	Reliable, industrial-grade
Honeywell	Micro switches (V-series)	USA-made, durable
Pepperl+Fuchs	Inductive proximity sensors (M12, M18)	German quality, expensive
Autonics	Photoelectric sensors, proximity switches	Korean brand, good value

K.8.2 Power Supplies

Vendor	Products	Notes
Mean Well	24V DC, 48V DC industrial supplies	Reliable, UL-listed, affordable
TDK-Lambda	High-power supplies (500W-3kW)	Industrial, very reliable
Phoenix Contact	DIN-rail power supplies	Compact, easy panel mounting

K.9 Tooling and Cutters

K.9.1 Endmills and Cutting Tools

Vendor	Products	Notes
Harvey Tool	Micro endmills, specialty tools	USA-made, excellent quality
OSG	Endmills, drills, taps	Japanese, carbide specialists
Kennametal	Indexable tools, inserts	Industrial, large-scale production
Onsrud	Router bits, wood tooling	CNC router specialist
Amana Tool	Router bits, compression bits	Wood and composite cutting

K.9.2 Budget Tooling

Vendor	Products	Notes
YG-1	Carbide endmills (1-20mm)	Korean, good quality/price ratio

Vendor	Products	Notes
ZCC-CT	Carbide tooling	Chinese, budget option for prototyping
Drill Hog	HSS drills, taps, bits	USA warehouse, fast shipping

K.10 Metrology and Measurement Tools

Vendor	Products	Notes
Mitutoyo	Calipers, micrometers, indicators	Japanese, industry standard
Starrett	Precision measuring tools	USA-made, premium quality
Fowler	Budget calipers, indicators	Good value, accurate
iGaging	Digital scales, DRO kits	Budget-friendly, hobbyist-grade
Renishaw	Touch probes, tool setters	Industrial, CNC automation

End of Vendor and Supplier Directory Appendix